

Weather Wise AI

Phase 3: Project Design

The design phase turns the requirements into an architecture, data flow and user interface plan.

System Architecture (layers)

Layer	Components
Presentation	Web/mobile UI: search bar, forecast dashboard, chat window, alerts page
Application	REST API server: auth, weather service, chat service, alert service
AI Layer	Forecast-improvement model + LLM assistant with prompt templates
Data Layer	Weather API connector, cache, database for users/preferences

Data Flow

- User enters a location or question in the UI.
- Backend fetches live and historical data (or reads from cache).
- ML model refines the forecast; the LLM turns data into a plain-language answer.
- Response and any alerts are returned and displayed to the user.

Database Design (main tables)

Table	Key fields
users	user_id, name, email, password_hash, created_at
locations	location_id, user_id, city, latitude, longitude
preferences	user_id, units, alert_types, alert_time
forecast_cache	location_id, data_json, fetched_at

UI Design Plan

- Home: current weather card, hourly strip, 7-day forecast.
- Chat: floating assistant for questions and advice.
- Alerts and Settings: manage locations, units and notifications.
- Design principles: clean layout, colour-coded conditions, dark/light mode.